

One Cup Chemistry

One Cup Chemistry

Student Lab Book v0.2

Predict • Measure • Explain • Challenge

Curriculum v0.2 - Avogadro recast

Moles → molecular scale → ions → matter → life → reasoning

How to Use This Book

1. Predict before measuring - Write what you think will happen and why. A prediction can be wrong and still be scientifically useful.

2. Preserve raw data - Record exactly what the instrument reports. Never “clean up” raw measurements to make a graph prettier.

3. Separate evidence types - Label statements as measured, calculated, simulated, inferred, or external evidence.

4. Use equations after the phenomenon - Equations are tools for compressing and testing observations, not substitutes for observing.

5. Challenge explanations - Every guide asks what assumption could be wrong and what evidence would change your mind.

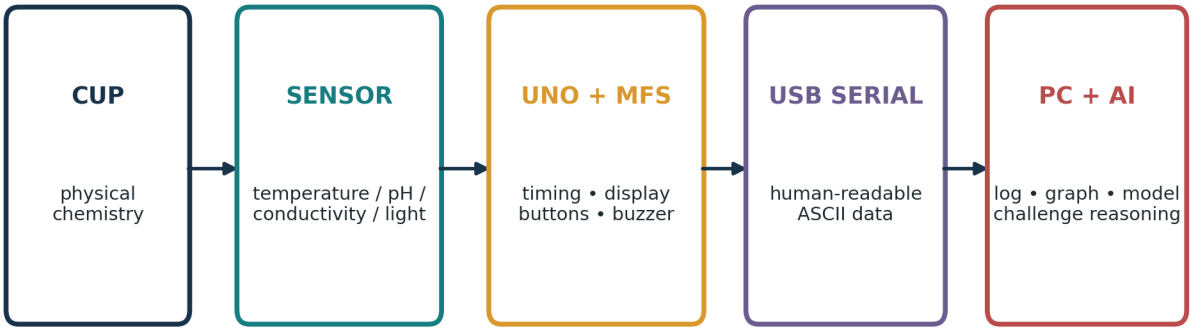
6. Treat AI as a reasoning participant - AI may predict, critique, calculate, or explain. Its answers are claims to be checked against the physical system.

Laboratory Rules

- Wear eye protection whenever the guide or teacher requires it.
- Nothing used in this laboratory is tasted or returned to food use.
- Keep liquids away from the UNO, USB cable, computer, and power connections.
- Use only low-voltage, current-limited electronics around solutions. Never connect mains electricity to electrodes.
- Commercial deicers are laboratory chemicals here; follow the product label and SDS.
- CaCl2 can heat strongly when dissolving. Add it gradually to cool/room-temperature water in an open vessel.
- Do not seal gas-producing reactions.
- Report spills, broken equipment, unexpected heating, or unusual odors immediately.

The Digital Bench

THE ONE CUP DATA SPINE



Physical data remain primary. AI is a reasoner over evidence, not a substitute for observation.

The cup stays understandable. The digital layer improves timing, repeatability, data storage, and the ability to compare a human explanation with an AI explanation.

Control	One Cup role
S1	Start/stop or accept
S2	Mark an event - salt added, stir, reagent added
S3	Change mode/sample
4-digit display	Current temperature, timer, or selected value
LEDs	Run/status/warning indicators
Buzzer	Hands-free confirmation
USB serial	Human-readable ASCII data to the PC

Human-readable data

```
META,OCC-05,RUN=3,SALT=CaCl2,MASS_WATER_G=100.0
T,0,22.84
MARK,5230,SALT_ADDED
T,6000,31.72
```

Your Evidence Vocabulary

MEASURED	Direct output from an instrument or direct observation.
CALCULATED	Produced from measured values using an explicit equation.
SIMULATED	Produced by a computational model.
INFERRED	An explanation or conclusion supported by evidence.
EXTERNAL EVIDENCE	A literature value, SDS, database structure, or other source outside this class measurement.

Rule: never let a calculation or AI explanation silently become “what we observed.”

Master Investigation Path

OCC-00 Measure Like a Chemist - Can two careful people measure the same thing and disagree?
OCC-01 Water as the First Standard - How close is 1 mL of water to 1 g in our room?
OCC-02 Newton Cooling - What shape does a cooling curve have?
OCC-03 Salt Changes the Curve - Does salt water cool differently from pure water?
OCC-04 The Heat-Capacity Trap - What does a fair comparison actually hold constant?
OCC-05 Heat of Dissolution - Why can adding a solid make water warmer or cooler?
OCC-06 Grams Are Not Moles - What does “the same amount” mean at the particle level?
OCC-07 Measuring a Molecule: Avogadro's Number from a Soap Film - Can a film one molecule thick let us count Avogadro's number?
OCC-08 Conductivity Reveals Ions - Why does salt water conduct while sugar water does not?
OCC-09 Freezing and Deicing - Why does road salt work?
OCC-10 From Saturated Solution to Crystal - How does an invisible solute become a visible crystal?
OCC-11 Water Inside a Solid: Hydrates - How can a solid crystal contain chemically organized water?
OCC-12 See the Lattice: Diffraction - How can a pattern of spots or peaks reveal an atomic lattice?
OCC-13 Acid and Base in the Cup - Can color reveal hydrogen-ion chemistry?
OCC-14 Make a Solid from Two Clear Solutions - Where did the precipitate come from?
OCC-15 Redox and Corrosion - Can a cup of salt water reveal electron transfer?
OCC-16 Rates and Catalysis - What changes how fast chemistry happens?
OCC-17 Light Becomes Concentration - Can a beam of light tell us how much solute is present?
OCC-18 From Ice Melt to Living Molecules - Why does biology care about hydrated Mg^{2+}?
OCC-19 Can the Reasoner Find the Flaw? - How intelligent is an AI when the experimental premise itself is wrong?

OCC-00

Measure Like a Chemist

Can two careful people measure the same thing and disagree?

PAGE A - PLAN + MEASURE

Mission

Can two careful people measure the same thing and disagree?

LEARNING TARGETS

- Distinguish resolution, precision, accuracy, and repeatability.
- Record raw data before calculating derived quantities.
- Estimate measurement uncertainty honestly.

MATERIALS

- Cup
- 0.1 g balance
- 100-250 mL graduated cylinder or syringe
- thermometer
- timer
- distilled water

Safety

Low hazard. Keep electronics dry and wipe spills promptly.

PREDICTION - BEFORE YOU TOUCH THE APPARATUS

Procedure

1. Zero/tare the balance with the empty cup.
2. Add a nominal 100 mL water by cylinder; record actual indicated volume.
3. Measure mass and temperature.
4. Repeat five times, emptying/drying as needed.
5. Compare class values and calculate mean, range, and optional standard deviation.

Raw data

Trial	Volume (mL)	Mass (g)	Temp (C)

OCC-00

Measure Like a Chemist

Can two careful people measure the same thing and disagree?

PAGE B - ANALYZE + EXPLAIN + CHALLENGE

UNO/MFS INSTRUMENTED MODE

MFS optional. Use buttons as event markers and the 4-digit display as a timer. This establishes the console before chemistry sensors are added.

Make meaning from the evidence

☐ Graph mass vs. trial.

☐ Identify which instrument dominates uncertainty.

☐ Write one sentence separating observation from interpretation.

GRAPH / CALCULATION / DIAGRAM SPACE

Theory - after the evidence

• A measurement is a number plus a unit plus an uncertainty.

• Random variation broadens repeated values; systematic error shifts them together.

• Density later links mass and volume, but first treat each instrument as an independent witness.

REASONING TRAP

"The display says 100.0" does not mean the delivered amount is exactly 100.0 mL.

AI REASONING CHALLENGE

Ask AI to critique the data table for false precision and missing metadata. Require it to identify what it cannot know from the numbers alone.

MY EXPLANATION - WHAT HAPPENED, AND WHAT EVIDENCE SUPPORTS IT?

WHAT EVIDENCE WOULD MAKE ME REVISE THIS EXPLANATION?

NEXT QUESTION **If mass and volume are both measurable, can water itself become a calibration material?**

OCC-01

Water as the First Standard

How close is 1 mL of water to 1 g in our room?

PAGE A - PLAN + MEASURE

Mission

How close is 1 mL of water to 1 g in our room?

LEARNING TARGETS

- Determine water density experimentally.
- Use a slope rather than a single ratio.
- Connect a macroscopic property to a reproducible material.

MATERIALS

- Cup
- balance
- graduated cylinder/syringes
- distilled water
- thermometer

Safety

Low hazard; keep balance pan dry.

PREDICTION - BEFORE YOU TOUCH THE APPARATUS

Procedure

1. Measure 25, 50, 75, 100, 125, and 150 mL portions into a tared cup.
2. Record total mass and temperature for each.
3. Plot mass vs. volume and fit a straight line.
4. Interpret slope as density.

Raw data

Volume (mL)	Mass (g)	Temp (C)	m/V

OCC-01

Water as the First Standard

How close is 1 mL of water to 1 g in our room?

PAGE B - ANALYZE + EXPLAIN + CHALLENGE

UNO/MFS INSTRUMENTED MODE

UNO/MFS can display water temperature during the density series. Log temperature so students can see that density claims always have conditions.

Make meaning from the evidence

☐ Fit m=aV+b. Discuss the meaning of nonzero b.

☐ Compare individual ratios with the fitted slope.

GRAPH / CALCULATION / DIAGRAM SPACE

Theory - after the evidence

• Density is ρ=m/V. For water near room temperature it is close to, but not exactly, 1 g/mL.

• A multi-point slope reduces the importance of a single reading error.

REASONING TRAP

"1 g = 1 mL" is a useful approximation, not an identity valid at every temperature.

AI REASONING CHALLENGE

Ask AI whether the intercept is evidence of "extra mass." Require at least three mundane instrumental explanations before any exotic one.

MY EXPLANATION - WHAT HAPPENED, AND WHAT EVIDENCE SUPPORTS IT?

WHAT EVIDENCE WOULD MAKE ME REVISE THIS EXPLANATION?

NEXT QUESTION **Now that we can measure temperature and time, what does a cup of hot water do while we simply watch it?**

OCC-02

Newton Cooling

What shape does a cooling curve have?

PAGE A - PLAN + MEASURE

Mission

What shape does a cooling curve have?

LEARNING TARGETS

- Collect a time series.
- Recognize exponential approach to ambient temperature.
- Separate temperature from heat.

MATERIALS

- Cup
- 150-250 g warm water
- temperature probe
- timer
- room-temperature probe optional
- lid with probe hole optional

Safety

Use warm water rather than scalding water. Teacher sets a maximum starting temperature appropriate to the room and vessel.

PREDICTION - BEFORE YOU TOUCH THE APPARATUS

Procedure

1. Record room temperature.
2. Prepare water at a convenient elevated temperature, e.g. 45-55 °C.
3. Start logging at fixed intervals for 15-30 min.
4. Do not move, stir, or relocate the cup during the run unless stirring is a defined condition.
5. Graph T vs. t and also T-Troom vs. t.

Raw data

Time	Sample T (C)	Room T (C)	Delta T

OCC-02

Newton Cooling

What shape does a cooling curve have?

PAGE B - ANALYZE + EXPLAIN + CHALLENGE

UNO/MFS INSTRUMENTED MODE

Default instrumented mode: DS18B20 on MFS A4, 1 sample/s or slower, S1 start/stop, S2 mark a disturbance, display current °C, USB serial log.

Make meaning from the evidence

☐ Estimate k by fitting T=Tenv+(T0-Tenv)e^(-kt).

☐ Compare two runs and discuss reproducibility.

GRAPH / CALCULATION / DIAGRAM SPACE

Theory - after the evidence

• Newton's law of cooling models dT/dt=-k(T-Tenv).

• The model is empirical and approximate: convection, evaporation, radiation, vessel heat capacity, and airflow all matter.

REASONING TRAP

An excellent mathematical fit does not prove the mechanism is complete.

AI REASONING CHALLENGE

Before giving the curve to AI, ask it to predict the qualitative shape and list assumptions. Afterward, ask which assumptions the data cannot test.

MY EXPLANATION - WHAT HAPPENED, AND WHAT EVIDENCE SUPPORTS IT?

WHAT EVIDENCE WOULD MAKE ME REVISE THIS EXPLANATION?

NEXT QUESTION **What happens if we keep the same water but dissolve salt in it?**

OCC-03

Salt Changes the Curve

Does salt water cool differently from pure water?

PAGE A - PLAN + MEASURE

Mission

Does salt water cool differently from pure water?

LEARNING TARGETS

- Prepare a reproducible NaCl solution.
- Compare two cooling curves under matched conditions.
- Identify variables unintentionally changed by adding solute.

MATERIALS

- Two identical cups
- 250 g water per trial
- NaCl (example 90 g for the deliberately naive condition)
- balance
- temperature logging

Safety

Ordinary NaCl is low hazard. Avoid ingestion because laboratory materials are never food after entering the lab workflow.

PREDICTION - BEFORE YOU TOUCH THE APPARATUS

Procedure

1. Run pure water as in OCC-02.
2. Prepare the second cup with the same starting water mass plus measured NaCl.
3. Dissolve completely; allow both samples to reach the same chosen starting temperature.
4. Run the cooling protocol with identical cup geometry and probe depth.

Raw data

Time	Water T	Brine T	Notes

OCC-03

Salt Changes the Curve

Does salt water cool differently from pure water?

PAGE B - ANALYZE + EXPLAIN + CHALLENGE

UNO/MFS INSTRUMENTED MODE

MFS produces identical time-stamped data format for water and brine. Use a run ID in the host logger so files can be overlaid automatically.

Make meaning from the evidence

☐ Record total sample mass.

☐ Calculate concentration by mass.

☐ List all variables changed when salt was added.

GRAPH / CALCULATION / DIAGRAM SPACE

Theory - after the evidence

• Salt changes solution properties including heat capacity, density, vapor pressure, and convection.

• But adding salt also changes total mass, so “lower specific heat” alone cannot predict total cooling behavior.

REASONING TRAP

The classic trap: comparing specific heat while ignoring m in Q=mcΔT.

AI REASONING CHALLENGE

Ask AI: “NaCl solution has lower specific heat than water. Which cools faster?” Do not mention the mass change. Score whether AI discovers the premise defect.

MY EXPLANATION - WHAT HAPPENED, AND WHAT EVIDENCE SUPPORTS IT?

WHAT EVIDENCE WOULD MAKE ME REVISE THIS EXPLANATION?

NEXT QUESTION **How do we redesign the comparison so “fair” has an explicit meaning?**

OCC-04

The Heat-Capacity Trap

What does a fair comparison actually hold constant?

PAGE A - PLAN + MEASURE

Mission

What does a fair comparison actually hold constant?

LEARNING TARGETS

- Distinguish specific heat c from total heat capacity C=mc.
- Design equal-total-mass and equal-water-mass comparisons.
- Understand that experimental fairness depends on the question.

MATERIALS

- Identical cups
- water
- NaCl
- balance
- temperature logger

Safety

Low hazard; standard hot-water precautions.

PREDICTION - BEFORE YOU TOUCH THE APPARATUS

Procedure

1. Design Condition A: equal starting water mass, then add salt.
2. Design Condition B: equal total solution mass, with one sample pure water and one brine.
3. Optionally design Condition C: equal moles of solvent.
4. Run matched cooling curves from the same starting temperature.

Raw data

Condition	Total mass	Start T	Cooling metric

OCC-04

The Heat-Capacity Trap

What does a fair comparison actually hold constant?

PAGE B - ANALYZE + EXPLAIN + CHALLENGE

UNO/MFS INSTRUMENTED MODE

Use MFS buttons to select condition A/B/C before each run; emit a header line such as META,MODE,EQUAL_MASS to preserve context in the data file.

Make meaning from the evidence

☐ Calculate approximate total thermal capacitance mc using reference specific heats if available.

☐ Explain which design answers which question.

GRAPH / CALCULATION / DIAGRAM SPACE

Theory - after the evidence

• Q=mcΔT. Lower c can be offset by larger m.

• There is no universal “fair test”; the controlled variables must correspond to the scientific question.

REASONING TRAP

“Control all variables” is impossible and often meaningless; control the variables required by the hypothesis.

AI REASONING CHALLENGE

Ask AI to propose a “fair” test. Reject answers that do not state what is held constant and why.

MY EXPLANATION - WHAT HAPPENED, AND WHAT EVIDENCE SUPPORTS IT?

WHAT EVIDENCE WOULD MAKE ME REVISE THIS EXPLANATION?

NEXT QUESTION **If different salts are used, does “same amount of salt” mean same grams or something else?**

OCC-05

Heat of Dissolution

Why can adding a solid make water warmer or cooler?

PAGE A - PLAN + MEASURE

Mission

Why can adding a solid make water warmer or cooler?

LEARNING TARGETS

- Observe endothermic/exothermic dissolution.
- Separate preparation heat from later cooling.
- Compare NaCl, $CaCl_2$, and $MgCl_2$-based deicer materials.

MATERIALS

- Three small cups
- 100 g water each
- 10-20 g NaCl
- 10-20 g $CaCl_2$ ice-melt pellets
- 10-20 g $MgCl_2$ hydrate product if available
- balance
- temperature probe

Safety

Goggles required. $CaCl_2$ dissolution can become hot; add small portions gradually to cool/room-temperature water and do not seal the vessel. Follow each product SDS.

PREDICTION - BEFORE YOU TOUCH THE APPARATUS

Procedure

1. Measure initial water temperature.
2. Add one salt gradually while stirring; log temperature continuously.
3. Repeat with fresh water for each salt and equalize the comparison basis.
4. After the dissolution event, allow samples to return to the same starting temperature before any cooling comparison.

Raw data

Time	Salt/sample	Temp (C)	Event

OCC-05

Heat of Dissolution

Why can adding a solid make water warmer or cooler?

PAGE B - ANALYZE + EXPLAIN + CHALLENGE

UNO/MFS INSTRUMENTED MODE

MFS shines here: 2-5 samples/s captures the transient; S2 marks the instant of addition; buzzer confirms event marking without taking eyes off the cup.

Make meaning from the evidence

☐ Measure ΔT from pre-addition baseline to peak/minimum.

☐ Compare energy per gram cautiously; rigorous molar enthalpy requires heat-capacity corrections.

GRAPH / CALCULATION / DIAGRAM SPACE

Theory - after the evidence

• Dissolution has an enthalpy change arising from lattice disruption and hydration of ions.

• Commercial deicers may be hydrates or blends, so composition must be documented.

REASONING TRAP

If you begin the cooling run immediately, you are mixing heat of dissolution with cooling behavior.

AI REASONING CHALLENGE

Ask AI to explain the transient. Require it to distinguish “the solution is hotter” from “the solution cools more slowly.”

MY EXPLANATION - WHAT HAPPENED, AND WHAT EVIDENCE SUPPORTS IT?

WHAT EVIDENCE WOULD MAKE ME REVISE THIS EXPLANATION?

NEXT QUESTION **Equal grams of NaCl, $CaCl_2$, and $MgCl_2$ are not equal chemically. What is the right counting unit?**

OCC-06

Grams Are Not Moles

What does “the same amount” mean at the particle level?

PAGE A - PLAN + MEASURE

Mission

What does “the same amount” mean at the particle level?

LEARNING TARGETS

- Use molar mass to convert grams to amount of substance.
- Compare equal-mass and equal-mole preparations.
- Recognize waters of hydration in formula mass.

MATERIALS

- NaCl
- $CaCl_2$
- $MgCl_2 \cdot 6H_2O$ or labeled $MgCl_2$ hydrate
- balance
- calculator
- water
- cups

Safety

Use the same PPE required by the salts. Do not assume commercial deicer purity; use label/SDS composition in calculations.

PREDICTION - BEFORE YOU TOUCH THE APPARATUS

Procedure

1. Calculate molar masses from formulas.
2. Determine moles in a fixed mass, e.g. 10.0 g.
3. Prepare equal-mole samples where feasible.
4. Record how much water of crystallization is carried by hydrated $MgCl_2$.

Raw data

Material	Mass (g)	Molar mass	Moles

OCC-06

Grams Are Not Moles

What does “the same amount” mean at the particle level?

PAGE B - ANALYZE + EXPLAIN + CHALLENGE

UNO/MFS INSTRUMENTED MODE

UNO/MFS is mostly a timer/display here; the host app can calculate recipe masses from a target molarity and display them before the run.

Make meaning from the evidence

☐ Compare moles of cations, chloride ions, and total ions for each preparation.

☐ State which basis is appropriate for the next experiment.

GRAPH / CALCULATION / DIAGRAM SPACE

Theory - after the evidence

• n=m/M. A hydrate’s formula mass includes its bound water.

• Equal grams can mean very different numbers of formula units and ions.

REASONING TRAP

Chemical formulas are counting statements, not just labels.

AI REASONING CHALLENGE

Ask AI to design “equal concentration” samples, then audit whether it used solution volume, solvent mass, or final mass consistently.

MY EXPLANATION - WHAT HAPPENED, AND WHAT EVIDENCE SUPPORTS IT?

WHAT EVIDENCE WOULD MAKE ME REVISE THIS EXPLANATION?

NEXT QUESTION A mole is a counting unit. Can ordinary geometry let us estimate how many molecules that means?

OCC-07

Measuring a Molecule: Avogadro's Number from a Soap Film

Can a film one molecule thick let us count Avogadro's number?

PAGE A - PLAN + MEASURE

Mission

Can a film one molecule thick let us count Avogadro's number?

LEARNING TARGETS

- Measure an oleic-acid monolayer.
- Estimate molecular length from film geometry.
- Estimate Avogadro's number from molecular volume.
- Separate measured quantities from model-based inference.

MATERIALS

- Oleic acid
- isopropyl alcohol dilution
- wide tray/cup of water
- fine powder
- calibrated dropper/syringe
- ruler/calipers

Safety

Yellow band. The alcohol diluent is flammable. Keep away from flames/heat, use ventilation, wear eye protection, and keep dilution labels with the sample.

PREDICTION - BEFORE YOU TOUCH THE APPARATUS

_____

_____

Procedure

1. Use the teacher-prepared dilution or record the exact dilution you prepare.
2. Calibrate the dropper by counting drops per 1.00 mL.
3. Dust the water surface lightly and evenly with powder.
4. Place one drop at the center and allow the clear monolayer region to finish spreading.
5. Measure the film diameter in two directions and average.
6. Repeat at least three trials.

Raw data | Stock dilution: _____ Final dilution: _____ Drops per 1.00 mL: _____

Trial	Diameter 1 (cm)	Diameter 2 (cm)	Average diameter (cm)

OCC-07

Measuring a Molecule: Avogadro's Number from a Soap Film

Can a film one molecule thick let us count Avogadro's number?

PAGE B - ANALYZE + EXPLAIN + CHALLENGE

UNO/MFS INSTRUMENTED MODE Use a fixed-position phone/camera image with a ruler in frame for repeatable diameter measurement. Advanced mode may use host image analysis; the cup remains the primary evidence source.

Make meaning from the evidence

☐ Calculate the pure oleic-acid volume delivered in one drop.

☐ Calculate $A = \pi r^2$, then estimate monolayer thickness $t = V/A$.

☐ Use a simple molecular-volume model to estimate Avogadro's number; compare order of magnitude with 6.022×10^{23}.

GRAPH / CALCULATION / DIAGRAM SPACE

Theory - after the evidence

• Oleic acid can form a one-molecule-thick monolayer on water. Film volume divided by area estimates molecular length; a simple molecular-volume model then links that scale to the mole.

REASONING TRAP

A monolayer and a simplified molecular shape are assumptions. “Close” means correct scale, not exact agreement.

AI REASONING CHALLENGE

Give AI the calibration, dilution, and diameter only. Require all assumptions and a sensitivity check before it reports N_A.

MY EXPLANATION - WHAT HAPPENED, AND WHAT EVIDENCE SUPPORTS IT?

WHAT EVIDENCE WOULD MAKE ME REVISE THIS EXPLANATION?

NEXT QUESTION We inferred molecular scale geometrically. Can charged particles in solution be detected electrically?

OCC-08

Conductivity Reveals Ions

Why does salt water conduct while sugar water does not?

PAGE A - PLAN + MEASURE

Mission

Why does salt water conduct while sugar water does not?

LEARNING TARGETS

- Distinguish dissolving from ion formation.
- Compare electrolytes and nonelectrolytes.
- Relate conductivity qualitatively to concentration and ion mobility.

MATERIALS

- Distilled water
- NaCl
- sucrose
- $CaCl_2$
- simple low-voltage conductivity probe or commercial module
- cups

Safety

Use only low-voltage sensor circuits. Do not use mains-connected electrodes. Rinse probe between samples.

PREDICTION - BEFORE YOU TOUCH THE APPARATUS

Procedure

1. Measure baseline distilled-water conductance.
2. Prepare matched mass or molar solutions of sugar and salts.
3. Measure each after rinsing the probe.
4. Make a dilution series for one salt.

Raw data

Sample	Concentration	Temp (C)	Conductivity

OCC-08

Conductivity Reveals Ions

Why does salt water conduct while sugar water does not?

PAGE B - ANALYZE + EXPLAIN + CHALLENGE

UNO/MFS INSTRUMENTED MODE

Analog conductivity module can use the free analog channel (often A5 on common MFS revisions) or an external ADC. Log temperature too because conductivity is temperature dependent.

Make meaning from the evidence

☐ Plot sensor response vs. concentration.

☐ Do not call the relationship universal unless calibrated for that probe and concentration range.

GRAPH / CALCULATION / DIAGRAM SPACE

Theory - after the evidence

• Ionic solids dissociate into mobile charge carriers; molecular solutes such as sucrose dissolve without producing comparable concentrations of ions.

• Conductivity depends on concentration, charge, mobility, geometry, and temperature.

REASONING TRAP

"Dissolved" and "ionized" are not synonyms.

AI REASONING CHALLENGE

Ask AI to predict the order of water, sugar, NaCl, $CaCl_2$. Then require a mechanistic explanation in terms of mobile charge carriers.

MY EXPLANATION - WHAT HAPPENED, AND WHAT EVIDENCE SUPPORTS IT?

WHAT EVIDENCE WOULD MAKE ME REVISE THIS EXPLANATION?

NEXT QUESTION **If ions change solution properties, why does salt melt ice?**

OCC-09

Freezing and Deicing

Why does road salt work?

PAGE A - PLAN + MEASURE

Mission

Why does road salt work?

LEARNING TARGETS

- Observe freezing-point depression qualitatively.
- Connect deicing to colligative behavior and phase equilibrium.
- Compare deicer formulations.

MATERIALS

- Small sealed sample tubes or cups
- water
- NaCl brine
- $CaCl_2$ brine
- ice-salt bath or freezer
- temperature probes

Safety

Cold-burn risk from very cold baths. Use gloves/tongs as appropriate. Keep glass out of student ice baths unless rated for thermal stress.

PREDICTION - BEFORE YOU TOUCH THE APPARATUS

Procedure

1. Prepare water and two brines of documented composition.
2. Cool under identical conditions while logging temperature.
3. Observe onset of freezing/slush and temperature plateaus.
4. Compare which sample remains liquid longest.

Raw data

Time	Pure water T	Brine T	Phase note

OCC-09

Freezing and Deicing

Why does road salt work?

PAGE B - ANALYZE + EXPLAIN + CHALLENGE

UNO/MFS INSTRUMENTED MODE

MFS can log one probe at a time; multiple samples can be run sequentially or with a small external sensor multiplexer in advanced mode.

Make meaning from the evidence

☐ Record concentration and approximate freezing onset.

☐ Discuss why $CaCl_2$ can be effective beyond NaCl under some conditions.

GRAPH / CALCULATION / DIAGRAM SPACE

Theory - after the evidence

• Dissolved particles lower the chemical potential of liquid water and depress the freezing point.

• The magnitude depends on particle concentration and non-ideal behavior; concentrated deicers are not ideal dilute solutions.

REASONING TRAP

Deicing performance is not determined by freezing point alone; dissolution heat, concentration, surface conditions, and practical formulation matter.

AI REASONING CHALLENGE

Ask AI to rank deicers, then require it to list at least three performance variables besides eutectic temperature.

MY EXPLANATION - WHAT HAPPENED, AND WHAT EVIDENCE SUPPORTS IT?

WHAT EVIDENCE WOULD MAKE ME REVISE THIS EXPLANATION?

NEXT QUESTION **When concentrated solutions lose water, where do the ions go?**

OCC-10

From Saturated Solution to Crystal

How does an invisible solute become a visible crystal?

PAGE A - PLAN + MEASURE

Mission

How does an invisible solute become a visible crystal?

LEARNING TARGETS

- Create saturated/supersaturated conditions.
- Observe nucleation and crystal growth.
- Relate solubility to temperature and evaporation.

MATERIALS

- NaCl or Epsom salt
- warm water
- cup
- balance
- evaporation dish or same cup
- thread/seed crystal optional

Safety

Use moderate water temperatures. Label solutions and keep them away from food preparation areas.

PREDICTION - BEFORE YOU TOUCH THE APPARATUS

Procedure

1. Make a near-saturated solution by adding solute until little more dissolves.
2. Filter/decant undissolved solid if needed.
3. Allow slow evaporation or controlled cooling.
4. Photograph crystals daily and record mass/size changes.

Raw data

Day/time	Solution mass	Crystal size	Observation

OCC-10

From Saturated Solution to Crystal

How does an invisible solute become a visible crystal?

PAGE B - ANALYZE + EXPLAIN + CHALLENGE

UNO/MFS INSTRUMENTED MODE

MFS can act as a long-duration room-temperature logger and audible reminder; host software can time-stamp daily images and observations.

Make meaning from the evidence

☐ Track mass loss from evaporation and crystal mass gained if separable.

☐ Compare slow vs. rapid growth conditions.

GRAPH / CALCULATION / DIAGRAM SPACE

Theory - after the evidence

• At saturation, dissolved and solid phases are at equilibrium. Supersaturation permits nucleation; crystal growth then lowers supersaturation.

REASONING TRAP

A crystal's macroscopic shape is not a simple magnified picture of one molecule.

AI REASONING CHALLENGE

Ask AI to infer lattice structure from crystal shape alone; then critique why that inference is limited.

MY EXPLANATION - WHAT HAPPENED, AND WHAT EVIDENCE SUPPORTS IT?

WHAT EVIDENCE WOULD MAKE ME REVISE THIS EXPLANATION?

NEXT QUESTION **Some salts contain water inside the solid itself. How can a "dry" crystal contain water?**

OCC-11

Water Inside a Solid: Hydrates

How can a solid crystal contain chemically organized water?

PAGE A - PLAN + MEASURE

Mission

How can a solid crystal contain chemically organized water?

LEARNING TARGETS

- Interpret hydrate formulas.
- Compare hydrated and anhydrous formula masses.
- Connect crystal water to coordination chemistry.

MATERIALS

- $MgCl_2 \cdot 6H_2O$ or $MgSO_4 \cdot 7H_2O$
- balance
- sealed storage
- optional gentle teacher demonstration of dehydration for a suitable hydrate

Safety

Hygroscopic salts absorb moisture. Heating/dehydration, if performed, is teacher controlled and compound-specific; consult SDS and avoid decomposition conditions.

PREDICTION - BEFORE YOU TOUCH THE APPARATUS

_____

_____

Procedure

1. Examine the hydrate formula and calculate the mass fraction of water.
2. Compare 10 g hydrate with 10 g nominal anhydrous salt on a mole basis.
3. Use a 3-D structure viewer for $MgCl_2 \cdot 6H_2O$.
4. Identify Mg, Cl, O, H and the repeating unit cell.

Raw data

Material/formula	Sample mass	Water fraction	Notes

OCC-11

Water Inside a Solid: Hydrates

How can a solid crystal contain chemically organized water?

PAGE B - ANALYZE + EXPLAIN + CHALLENGE

UNO/MFS INSTRUMENTED MODE

Use MFS as a temperature/humidity companion if an external humidity sensor is added; otherwise keep the digital thread by recording sample IDs and masses through the host.

Make meaning from the evidence

☐ Calculate percent water by mass.

☐ Explain why equal grams of hydrate and anhydrous material produce different solution compositions.

GRAPH / CALCULATION / DIAGRAM SPACE

Theory - after the evidence

• Hydrate water occupies defined crystallographic positions; it is not simply liquid trapped between grains.

• Mg^{2+} commonly exhibits sixfold water coordination in the hexahydrate structure.

REASONING TRAP

Dot notation in a hydrate formula is structural stoichiometry, not multiplication by “wetness.”

AI REASONING CHALLENGE

Ask AI to draw or describe $MgCl_2 \cdot 6H_2O$. Then compare its model with the actual crystallographic structure rather than accepting a plausible cartoon.

MY EXPLANATION - WHAT HAPPENED, AND WHAT EVIDENCE SUPPORTS IT?

WHAT EVIDENCE WOULD MAKE ME REVISE THIS EXPLANATION?

NEXT QUESTION **How do we know the atoms really occupy those positions?**

OCC-12

See the Lattice: Diffraction

How can a pattern of spots or peaks reveal an atomic lattice?

PAGE A - PLAN + MEASURE

Mission

How can a pattern of spots or peaks reveal an atomic lattice?

LEARNING TARGETS

- Load a real CIF structure.
- Connect unit cells to diffraction.
- Understand Bragg's law conceptually.

MATERIALS

- Internet-connected computer
- COD $MgCl_2 \cdot 6H_2O$ entry 9011352
- OpenMX Viewer or equivalent
- XRD simulation tool

Safety

No chemical hazard if conducted as a digital extension. Actual X-ray equipment is not part of this curriculum and requires specialized safety infrastructure.

PREDICTION - BEFORE YOU TOUCH THE APPARATUS

_____

_____

Procedure

1. Open COD 9011352 in a browser-based structure viewer.
2. Display the unit cell and expand to a 2×2×2 supercell.
3. Identify the Mg coordination environment and chloride positions.
4. Generate or view a simulated powder diffraction pattern.
5. Relate peak positions qualitatively to lattice spacings using Bragg's law.

Raw data

View/pattern	Feature	Evidence type	Interpretation

OCC-12

See the Lattice: Diffraction

How can a pattern of spots or peaks reveal an atomic lattice?

PAGE B - ANALYZE + EXPLAIN + CHALLENGE

UNO/MFS INSTRUMENTED MODE

Host/AI mode: load CIF metadata alongside lab records so students can ask structural questions grounded in the actual file, not generic chemistry.

Make meaning from the evidence

☐ Choose one prominent peak and explain what information its angle carries.

☐ Distinguish structure solution from simply viewing a CIF.

GRAPH / CALCULATION / DIAGRAM SPACE

Theory - after the evidence

• Periodic arrays scatter X-rays coherently. Bragg's law nλ=2d sinθ links wavelength, plane spacing, and diffraction angle.

• X-rays locate electron density well; neutron diffraction is especially useful for hydrogen positions.

REASONING TRAP

A diffraction pattern is not a photograph of atoms; structure is inferred from reciprocal-space measurements and models.

AI REASONING CHALLENGE

Ask AI to identify the unit-cell parameters and Mg coordination from the CIF. Verify every answer against the source record.

MY EXPLANATION - WHAT HAPPENED, AND WHAT EVIDENCE SUPPORTS IT?

WHAT EVIDENCE WOULD MAKE ME REVISE THIS EXPLANATION?

NEXT QUESTION **We have explored neutral water and salts. What changes when the solution becomes acidic or basic?**

OCC-13

Acid and Base in the Cup

Can color reveal hydrogen-ion chemistry?

PAGE A - PLAN + MEASURE

Mission

Can color reveal hydrogen-ion chemistry?

LEARNING TARGETS

- Use an indicator to classify acids and bases.
- Connect pH to a logarithmic concentration scale.
- Observe neutralization.

MATERIALS

- Red cabbage indicator or pH paper
- vinegar/citric acid
- baking soda solution
- washing soda solution optional
- droppers
- cups

Safety

Household dilute reagents are low hazard but goggles remain standard. Never mix unknown cleaners or bleach with acids.

PREDICTION - BEFORE YOU TOUCH THE APPARATUS

Procedure

1. Prepare indicator and a neutral water reference.
2. Test small equal volumes of acidic and basic household solutions.
3. Create a stepwise neutralization by adding base to acid in small measured increments.
4. Record color/pH after each increment.

Raw data

Addition #	Added volume	pH/color	Observation

OCC-13

Acid and Base in the Cup

Can color reveal hydrogen-ion chemistry?

PAGE B - ANALYZE + EXPLAIN + CHALLENGE

UNO/MFS INSTRUMENTED MODE

A pH module can use the available analog channel; log pH and temperature. Buttons mark each addition so the titration trace contains exact event times.

Make meaning from the evidence

☐ Plot pH vs. added volume if a pH probe is available.

☐ Explain why equal household volumes do not necessarily neutralize.

GRAPH / CALCULATION / DIAGRAM SPACE

Theory - after the evidence

• Acid-base chemistry concerns proton transfer and solution equilibria. pH=-log10[H^+] in the idealized introductory definition.

REASONING TRAP

Color is evidence of indicator equilibrium, not a direct count of “acid particles.”

AI REASONING CHALLENGE

Ask AI to predict how many mL of base will neutralize vinegar using only labels. It should refuse overprecision without concentration data.

MY EXPLANATION - WHAT HAPPENED, AND WHAT EVIDENCE SUPPORTS IT?

WHAT EVIDENCE WOULD MAKE ME REVISE THIS EXPLANATION?

NEXT QUESTION **What happens when two clear ionic solutions create an insoluble solid?**

OCC-14

Make a Solid from Two Clear Solutions

Where did the precipitate come from?

PAGE A - PLAN + MEASURE

Mission

Where did the precipitate come from?

LEARNING TARGETS

- Observe precipitation.
- Write complete and net ionic equations.
- Introduce solubility equilibrium and spectator ions.

MATERIALS

- Dilute $CaCl_2$ solution
- dilute Na_2CO_3 solution (washing soda)
- droppers/cylinders
- cup
- filter paper optional

Safety

Goggles. Use dilute solutions. Dispose according to local classroom rules; avoid drain disposal of any extension using heavy-metal salts.

PREDICTION - BEFORE YOU TOUCH THE APPARATUS

Procedure

1. Prepare known dilute solutions, e.g. 0.1 M if quantitative work is desired.
2. Mix equal measured volumes.
3. Observe turbidity/solid formation.
4. Filter, rinse, dry, and weigh precipitate in an advanced mass-balance version.

Raw data

Trial	Vol A	Vol B	Precipitate/result

OCC-14

Make a Solid from Two Clear Solutions

Where did the precipitate come from?

PAGE B - ANALYZE + EXPLAIN + CHALLENGE

UNO/MFS INSTRUMENTED MODE

Use an LED/light sensor across the cup as a crude turbidity channel; time the onset and clearing/settling behavior.

Make meaning from the evidence

☐ Predict limiting reagent from moles.

☐ Compare theoretical and recovered mass if dried quantitatively.

GRAPH / CALCULATION / DIAGRAM SPACE

Theory - after the evidence

- $Ca^{2+} + CO_3^{2-} \rightarrow CaCO_3(s)$.
- Na^+ and Cl^- remain in solution as spectator ions.
- Precipitation reflects a solubility equilibrium, not disappearance of ions.

REASONING TRAP

Cloudiness is not “a new element”; it is a new phase with atoms conserved.

AI REASONING CHALLENGE

Ask AI to balance molecular and net ionic equations and to identify spectator ions. Verify charge and atom conservation.

MY EXPLANATION - WHAT HAPPENED, AND WHAT EVIDENCE SUPPORTS IT?

WHAT EVIDENCE WOULD MAKE ME REVISE THIS EXPLANATION?

NEXT QUESTION **Ions can also transfer electrons through metals. Can the same cup become an electrochemical cell?**

OCC-15

Redox and Corrosion

Can a cup of salt water reveal electron transfer?

PAGE A - PLAN + MEASURE

Mission

Can a cup of salt water reveal electron transfer?

LEARNING TARGETS

- Recognize oxidation and reduction.
- Measure voltage between dissimilar metals.
- Connect corrosion to electrochemical processes.

MATERIALS

- NaCl solution
- copper strip
- zinc or galvanized steel strip
- iron nail
- multimeter
- alligator leads

Safety

Use only passive galvanic measurements. Do not connect mains power or perform uncontrolled electrolysis. Wash hands after handling metals.

PREDICTION - BEFORE YOU TOUCH THE APPARATUS

Procedure

1. Place two dissimilar clean metal strips in the electrolyte without touching.
2. Measure open-circuit voltage and polarity.
3. Try different pairs.
4. Observe surfaces over hours/days for corrosion products.

Raw data

Metal pair	Electrolyte	Voltage	Surface change

OCC-15

Redox and Corrosion

Can a cup of salt water reveal electron transfer?

PAGE B - ANALYZE + EXPLAIN + CHALLENGE

UNO/MFS INSTRUMENTED MODE

UNO can read a safely scaled low voltage through an analog input, but the handheld multimeter is preferred for first exposure. MFS remains timer/event logger.

Make meaning from the evidence

☐ Record sign as well as magnitude.

☐ Relate which electrode is negative to likely oxidation in the simple cell.

GRAPH / CALCULATION / DIAGRAM SPACE

Theory - after the evidence

• Oxidation releases electrons; reduction consumes them. Dissimilar electrodes in an electrolyte can establish a potential difference.

• Corrosion is coupled electrochemistry influenced strongly by chloride.

REASONING TRAP

A voltmeter reading alone does not identify every reaction occurring at the surfaces.

AI REASONING CHALLENGE

Ask AI to predict electrode polarity, then compare with the measured sign. Require it to explain discrepancies rather than flip labels silently.

MY EXPLANATION - WHAT HAPPENED, AND WHAT EVIDENCE SUPPORTS IT?

WHAT EVIDENCE WOULD MAKE ME REVISE THIS EXPLANATION?

NEXT QUESTION **Chemical change can be fast or slow. What controls reaction rate?**

OCC-16

Rates and Catalysis

What changes how fast chemistry happens?

PAGE A - PLAN + MEASURE

Mission

What changes how fast chemistry happens?

LEARNING TARGETS

- Measure a reaction rate proxy.
- Test concentration or temperature as a variable.
- Observe catalysis using a familiar biological catalyst.

MATERIALS

- 3% hydrogen peroxide
- dry yeast suspension (catalase source)
- small cup
- temperature probe
- graduated cylinder
- dish soap optional for visible foam

Safety

Goggles required. Use only household 3% H_2O_2. Keep volumes small; reaction foams and releases oxygen. Do not seal the vessel.

PREDICTION - BEFORE YOU TOUCH THE APPARATUS

_____

_____

Procedure

1. Prepare equal H_2O_2 portions at two temperatures or concentrations.
2. Add equal yeast suspension and start timing immediately.
3. Measure foam height, gas volume proxy, mass loss, or time to a defined endpoint.
4. Repeat with one variable changed at a time.

Raw data

Time	Condition A	Condition B	Rate metric

OCC-16

Rates and Catalysis

What changes how fast chemistry happens?

PAGE B - ANALYZE + EXPLAIN + CHALLENGE

UNO/MFS INSTRUMENTED MODE

MFS timer plus event button gives repeatable start marks. A pressure/gas sensor can be added in advanced mode, but never seal a peroxide reaction without engineered pressure relief.

Make meaning from the evidence

☐ Choose one rate metric before seeing the result.

☐ Graph metric vs. time and compare initial slopes where meaningful.

GRAPH / CALCULATION / DIAGRAM SPACE

Theory - after the evidence

• Catalysts lower the kinetic barrier without being consumed stoichiometrically. Reaction rate depends on concentration, temperature, catalysts, and mechanism.

REASONING TRAP

Foam height is a proxy influenced by surfactant and bubble stability; it is not a direct oxygen meter.

AI REASONING CHALLENGE

Ask AI to propose the cleanest rate metric using only available equipment. Score it on measurability and confounding variables.

MY EXPLANATION - WHAT HAPPENED, AND WHAT EVIDENCE SUPPORTS IT?

WHAT EVIDENCE WOULD MAKE ME REVISE THIS EXPLANATION?

NEXT QUESTION **Can light become a quantitative chemical instrument?**

OCC-17

Light Becomes Concentration

Can a beam of light tell us how much solute is present?

PAGE A - PLAN + MEASURE

Mission

Can a beam of light tell us how much solute is present?

LEARNING TARGETS

- Build a calibration curve.
- Introduce transmittance and absorbance.
- Use Beer-Lambert behavior over a suitable range.

MATERIALS

- Food coloring
- water
- identical clear cuvettes/cups or fixed optical path
- LED
- phototransistor/LDR
- resistors
- balance or syringes

Safety

Use low-voltage LEDs only. Avoid staring into high-intensity sources; ordinary visible indicator LEDs are sufficient.

PREDICTION - BEFORE YOU TOUCH THE APPARATUS

Procedure

1. Prepare a serial dilution of one food dye.
2. Keep path length and detector geometry fixed.
3. Measure detector response for blank and each concentration.
4. Convert to relative transmittance; optionally compute A=-log10(T).
5. Plot absorbance vs. concentration.

Raw data

Standard	Concentration	Detector	Absorbance

OCC-17

Light Becomes Concentration

Can a beam of light tell us how much solute is present?

PAGE B - ANALYZE + EXPLAIN + CHALLENGE

UNO/MFS INSTRUMENTED MODE

Photodetector can use A5 on common MFS boards when not occupied by another add-on. MFS buttons step through standards; USB stream labels each reading with concentration ID.

Make meaning from the evidence

☐ Build a calibration line, then estimate an unknown concentration.

☐ Report the range over which calibration is credible.

GRAPH / CALCULATION / DIAGRAM SPACE

Theory - after the evidence

• Beer-Lambert law A=εbc predicts absorbance proportional to concentration for suitable dilute systems, fixed wavelength, and fixed path length.

REASONING TRAP

A pretty straight line outside the instrument's useful range is not evidence of unlimited linearity.

AI REASONING CHALLENGE

Ask AI to infer the unknown from calibration data, then require uncertainty and detection-range caveats.

MY EXPLANATION - WHAT HAPPENED, AND WHAT EVIDENCE SUPPORTS IT?

WHAT EVIDENCE WOULD MAKE ME REVISE THIS EXPLANATION?

NEXT QUESTION How does the magnesium ion first encountered as ice melt connect to living chemistry?

OCC-18

From Ice Melt to Living Molecules

Why does biology care about hydrated Mg^{2+}?

PAGE A - PLAN + MEASURE

Mission

Why does biology care about hydrated Mg^{2+}?

LEARNING TARGETS

- Connect inorganic hydration to biochemistry.
- Explain Mg^{2+} roles in ATP and nucleic-acid chemistry.
- Distinguish a conceptual biological extension from a kitchen-table biochemical assay.

MATERIALS

- $MgCl_2 \cdot 6H_2O$ structure model/CIF
- molecular visualization resources
- ATP/DNA polymerase diagrams or reputable references
- student lab notebook

Safety

This is primarily a model-and-evidence lesson. Do not improvise biological or diagnostic experiments with deicer-grade material.

PREDICTION - BEFORE YOU TOUCH THE APPARATUS

_____

_____

Procedure

1. Return to the $MgCl_2 \cdot 6H_2O$ lattice and identify the hydrated Mg^{2+} environment.
2. Trace the conceptual path: hydrated Mg^{2+} → phosphate interactions → Mg-ATP → enzymes → DNA/RNA.
3. Examine a reputable PCR reference showing Mg^{2+} as a DNA-polymerase cofactor.
4. Construct an evidence map separating what was measured in class from what comes from structural/biochemical literature.

Raw data

Claim	Evidence source	Evidence type	Confidence

OCC-18

From Ice Melt to Living Molecules

Why does biology care about hydrated Mg^{2+}?

PAGE B - ANALYZE + EXPLAIN + CHALLENGE

UNO/MFS INSTRUMENTED MODE

The host/AI layer can now query the entire evidence chain: run data, formulas, CIF metadata, and curated references. The cup remains the physical anchor.

Make meaning from the evidence

☐ Create a concept map with citations.

☐ Label each arrow as observed, calculated, simulated, or literature-supported.

GRAPH / CALCULATION / DIAGRAM SPACE

Theory - after the evidence

• Mg^{2+} has strong hydration chemistry and is a ubiquitous biological cofactor. In PCR, Mg^{2+} is supplied commonly as $MgCl_2$ and supports DNA polymerase chemistry and charge stabilization.

REASONING TRAP

Conceptual continuity is not proof that a kitchen deicer sample can be used in biological systems. Purity and context matter.

AI REASONING CHALLENGE

Ask AI to explain the entire chain in 200 words, then force it to tag every claim as “our measurement,” “calculation,” “simulation,” or “external evidence.”

MY EXPLANATION - WHAT HAPPENED, AND WHAT EVIDENCE SUPPORTS IT?

WHAT EVIDENCE WOULD MAKE ME REVISE THIS EXPLANATION?

NEXT QUESTION **Can an AI reason across the whole chain without smuggling in assumptions?**

OCC-19

Can the Reasoner Find the Flaw?

How intelligent is an AI when the experimental premise itself is wrong?

PAGE A - PLAN + MEASURE

Mission

How intelligent is an AI when the experimental premise itself is wrong?

LEARNING TARGETS

- Treat AI output as testable scientific reasoning.
- Design prompts that expose assumptions.
- Use physical data to audit explanations.

MATERIALS

- Any completed OCC dataset
- PC with AI access
- student prompt log
- rubric

Safety

No additional chemical hazard. Privacy: do not send student personal data or restricted information to external systems.

PREDICTION - BEFORE YOU TOUCH THE APPARATUS

_____

_____

Procedure

1. Choose one prior experiment with a hidden or tempting assumption.
2. Before showing data, ask AI for a prediction and required assumptions.
3. Provide the actual protocol and ask for a critique.
4. Provide data and request an explanation.
5. Score whether it notices mass, concentration, hydrate state, sensor limits, or other premise issues before being prompted.

Raw data

AI stage	Claim	Assumption found?	Evidence test

OCC-19

Can the Reasoner Find the Flaw?

How intelligent is an AI when the experimental premise itself is wrong?

PAGE B - ANALYZE + EXPLAIN + CHALLENGE

UNO/MFS INSTRUMENTED MODE

UNO/MFS + host bridge makes the test unusually clean: the same machine-readable data stream can be inspected by humans, plotted by software, and presented to the AI without manual transcription.

Make meaning from the evidence

☐ Use a rubric: assumption detection, physical correctness, quantitative consistency, uncertainty, falsifiability, and revision quality.

GRAPH / CALCULATION / DIAGRAM SPACE

Theory - after the evidence

- Scientific intelligence is not just calculation. It includes premise checking, model selection, uncertainty, and willingness to revise.
- AI should be evaluated against physical evidence and explicit reasoning criteria.

REASONING TRAP

The goal is not to “catch the AI being wrong.” The goal is to distinguish fluent completion from disciplined scientific reasoning.

AI REASONING CHALLENGE

Ask the AI to state what observation would falsify its explanation. A reasoner that cannot name disconfirming evidence has not finished the scientific job.

MY EXPLANATION - WHAT HAPPENED, AND WHAT EVIDENCE SUPPORTS IT?

WHAT EVIDENCE WOULD MAKE ME REVISE THIS EXPLANATION?

NEXT QUESTION **The curriculum loops back to OCC-00: measurement, evidence, uncertainty, and disciplined inference.**

Final Evidence Audit

Choose any claim from OCC-00 through OCC-19 and reconstruct its evidence chain. Your final explanation should make clear what you personally measured, what you calculated, what a model simulated, what you inferred, and what came from an external source.

CLAIM

MEASURED EVIDENCE

CALCULATION/MODEL

EXTERNAL EVIDENCE

ALTERNATIVE EXPLANATION

OBSERVATION THAT COULD FALSIFY MY EXPLANATION

ONE CUP CHEMISTRY

The laboratory did not become sophisticated because the equipment became sophisticated. The questions became sophisticated.